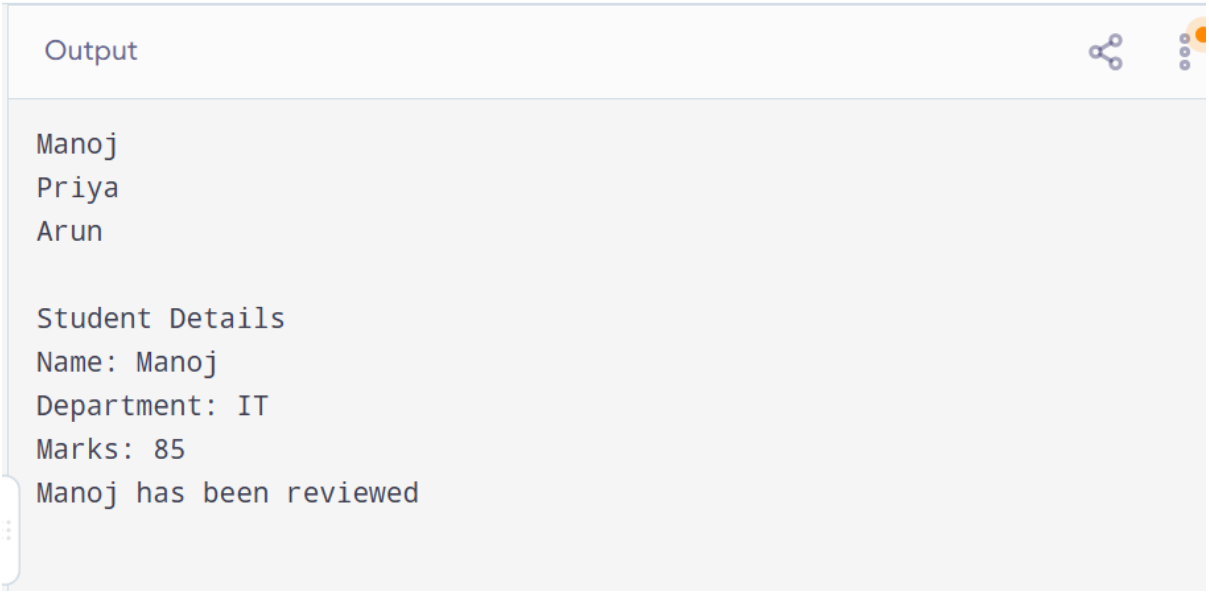
Task 1: A List That Opens a Detail Screen

```
import Foundation
struct Student {
    let name: String
    let department: String
    let marks: Int
}
let students = [
    Student(name: "Manoj", department: "IT", marks: 85),
    Student(name: "Priya", department: "CSE", marks: 90),
    Student(name: "Arun", department: "ECE", marks: 78)
]
func showDetails(_ student: Student) {
    print("\nStudent Details")
    print("Name: \(student.name)")
    print("Department: \(student.department)")
    print("Marks: \(student.marks)")
}

func markReviewed(_ student: Student, completion: (String) -> Void) {
    completion("\(student.name) has been reviewed")
}
for student in students {
    print(student.name)
}
showDetails(students[0])

markReviewed(students[0]) { message in
    print(message)
}
```



Task 2: Draggable and Pinchable Box

```
import Foundation
struct Box {
    var x: Double = 0
    var y: Double = 0
    var size: Double = 100
    mutating func drag(x: Double, y: Double) {
        self.x += x
        self.y += y
    }
    mutating func pinch(scale: Double) {
        size *= scale
    }
    mutating func reset() {
        x = 0
        y = 0
        size = 100
    }
```

```
    func show() {
        print("Position: (\(x), \(y))")
        print("Size: \(size) x \(size)")
    }
}

var box = Box()

print("Original box:")
box.show()

box.drag(x: 50, y: 30)
print("\nAfter dragging:")
box.show()

box.pinch(scale: 1.5)
print("\nAfter pinch:")
box.show()

box.reset()
print("\nAfter double tap:")
box.show()
```

Output

```
Original box:
Position: (0.0, 0.0)
Size: 100.0 x 100.0

After dragging:
Position: (50.0, 30.0)
Size: 100.0 x 100.0

After pinch:
Position: (50.0, 30.0)
Size: 150.0 x 150.0

After double tap:
Position: (0.0, 0.0)
Size: 100.0 x 100.0
```

Task 3: Segues and Unwind

```
import Foundation
class FirstScreen {
    var studentName = "Manoj"
    func openSecondScreen() -> SecondScreen {
        print("Opening second screen")
        return SecondScreen(name: studentName)
    }
    func unwind(message: String) {
        print("Returned to first screen")
        print("Result: \(message)")
    }
}
class SecondScreen {
    let name: String
```

```
    init(name: String) {
        self.name = name
    }

    func showDetails() {
        print("Student Name: \(name)")
    }

    func done(completion: (String) -> Void) {
        completion("\(name) completed the task")
    }
}
let first = FirstScreen()
let second = first.openSecondScreen()

second.showDetails()

second.done { result in
    first.unwind(message: result)
}
```

Output

```
Opening second screen
Student Name: Manoj
Returned to first screen
Result: Manoj completed the task
```